Harnessing Machine Learning for Urban Air Mobility: Insights from Air Taxi Communication, Demand Prediction, and Navigation Applications

AJUNA P. JOHN[a]

[a] *Purdue University, 550 Stadium Mall Dr., W. Lafayette, IN 47907, USA*

ABSTRACT

Urban Air Mobility (UAM) is emerging as a groundbreaking solution to the growing challenges of urban transportation. It leverages machine learning (ML) technologies to improve efficiency, safety, and sustainability. This literature review paper explores the latest advancements in ML applications within the UAM domain, focusing on three key application areas: demand prediction for air taxi services, autonomous navigation in complex urban environments, and communication systems for UAM vehicles. In the context of air taxi services, a major use case of UAM applications, the ML algorithms are essential for predicting passenger demand, which helps optimize resource allocation and operational strategies. Accurate demand forecasting ensures that UAM services can effectively meet fluctuating urban transportation needs. For autonomous navigation, ML—specifically, deep reinforcement learning—enables UAM vehicles to navigate urban landscapes efficiently by optimizing flight trajectories. This minimizes energy consumption and reduces noise pollution, enhancing the overall sustainability of UAM operations. Moreover, ML is critical in improving UAM communication systems by addressing challenges related to reliable data transmission and real-time decision-making. These communication advancements are crucial for coordinating UAM movements and ensuring safe and efficient operations in densely populated areas. By synthesizing insights from various journal articles and technical reports, this review highlights the transformative potential of ML in advancing UAM technologies. The integration of ML in these areas holds promise for creating more adaptive, sustainable, and efficient air mobility solutions, fundamentally reshaping the future of urban transportation and offering new possibilities for addressing congestion and environmental concerns.

1. Introduction

Urban Air Mobility (UAM) is a fast-growing solution to urban transportation challenges. UAM is only one of several pillars of emerging transportation technologies (ETT) and machine learning is only one of the several elements within each pillar (Mahdavian et al, 2021; Labi et al., 2021; Wei et al., 2022; Labi and Sinha, 2022; Huang et al., 2023; Milakis and Seibert, 2024; Rieger et al., 2024). In documenting the use of machine learning (ML) for urban air mobility, this paper addresses only one specific pillar of emerging transport technology (airborne vehicles) and within this pillar, two elements (the enabling technology of ML and the planning tool of demand estimation). Other ETT pillars include vehicle autonomy (Mahmassani, 2016), vehicle connectivity (Lu et al., 2014), shared transportation including micromobility (Feng et al., 2021), and electric propulsion (Carlsson & Olof 2003; Patel et al., 2021; Pamidimukkala et al., 2024). Other sub-elements of the "enabling technologies" element in air mobility and other pillars include geo-analytics (Ang et al., 2022) and AI (Dong et al., 2022; Brena, 2025; Gronfula and Sayed, 2025; Dong et al., 2023).

Other elements within each pillar include human factors (Scheff et al., 2020; Chauhan and Carroll, 2021; Li et al., 2023; Grindley et al., 2024), infrastructure (Guo et al., 2022; Kennard, 2023; Labi et al., 2023; Carrera et al., 2025), operational control (Marzbani et al., 2019; Chen et al., 2021; Du et al., 2022; Abdallaoui et al., 2023; Reda et al., 2024; Li et al., 2024), and deployment (Dai et al., 2025; Charnsethikul et al., 2025; Di Vito et al., 2023). Some studies have discussed the relationships between the pillars, specifically, how the co-existence of two pillars enhances (or degrades) the performance of at least one of the pillars (Michel et al., 2016; Anderson et al., 2016; Lin et al., 2017; Ha et al., 2021; Pourgholamali et al., 2023).

Machine learning has been applied extensively in several pillars (including autonomous vehicles) and applications (including system operations). For example, Bharilyam and Kumar (2024) used machine learning for trajectory prediction, Hassan et al. (2025) surveyed the machine learning landscape in autonomous vehicles. In the vehicle autonomy pillar, machine learning has been used for operations tasks including object classification (Alqarqaz et al., 2023), cooperative perception (Yang et al., 2021), and tactical lane-changing control (Dong et al., 2021).

This paper addresses three application areas of machine learning in urban air mobility: communication, demand prediction, and navigation. Other application areas of machine learning in urban air mobility include repositioning of air taxis within a region as has been done for micromobility devices within a city using other tools (Feng et al., 2021). Air taxi landing pad location planning and associated user pricing will be an issue as is the case for autonomous vehicles (Labi et al., 2023). Also, air taxis operations will use "highways in the sky" that is, air traffic lanes, which are virtual, structured 3D tubular spaces designed (Williams, 2000; Bauranov & Rakas, 2021), will need to resolve issues including relative localization and "lane" management (Marzouk, 2022) similar to highway lanes which are 2D (Seilabi et al., 2020; Khattak et al., 2022; Ji et al., 2025).

This paper discusses cases in the literature where machine learning (ML), as an enabling technology, has been used to improve the efficiency, safety, and sustainability of UAMs. The paper represents a synthesis of the literature on this subject, exploring the latest advancements in ML applications in the UAM domain. The review focuses on three key areas: communication systems for UAM vehicles, autonomous navigation in complex urban environments, and demand prediction for air taxi services.

2. Communication Systems for UAM Vehicles

Urban Air Mobility (UAM) and its subdomain of air taxi communication systems are expected to transform urban transportation by enabling autonomous, efficient, and safe air travel. However, integrating these systems into existing urban infrastructure poses unique challenges, particularly in communication reliability, latency, and security. Machine learning (ML) has become instrumental in addressing these issues. The following literature review synthesizes research on machine learning techniques applied to predictive communication management, performance optimization, and secure communication in UAM systems.

(a). Predictive Handover in Unmanned Aerial Systems (UAS)

Aydin and Rodosek (2023) investigate the use of ML for predictive handover in unmanned aerial systems (UAS) communication networks, which share operational characteristics with UAM systems in urban air taxis. They highlight that the continuous movement of UAS in varying urban environments requires seamless handovers between communication networks to ensure reliable connectivity. This challenge is heightened in urban areas due to dense buildings and frequent signal obstructions. The researchers proposed machine-learning models that predict optimal handover points based on environmental and mobility data, thereby minimizing connectivity losses. This approach is beneficial for UAM air taxis, which require robust connectivity to support navigation and control systems in densely populated regions.

(b). GNSS Performance Prediction Using Environment Recognition

Isik et al. (2023) explored ML-based Global Navigation Satellite System (GNSS) performance prediction in urban settings, which can be particularly valuable for UAM systems operating in areas with complex geographical and infrastructural layouts. They use environment recognition to address disruptions in GNSS signals in urban canyons caused by signal multipath effects and obstructions. By training a model on environmental features, such as building density and height, their system can predict GNSS performance and switch to alternative navigation aids if necessary. This capability is considered crucial for UAM air taxis to ensure consistent navigation accuracy, particularly during takeoff, landing, and low-altitude maneuvers in densely populated urban areas.

(c). Machine Learning Techniques for Emergency Situations in Air Mobility

Zhou, Liu, and Song (2023) surveyed machine learning algorithms applicable to air mobility in emergency scenarios, focusing on ML models designed for real-time decision making under unpredictable conditions. These algorithms include reinforcement learning for optimizing response time and reducing risks in emergencies, such as sudden communication link failures or adverse weather conditions. Such techniques are essential for UAM systems, where communication interruptions or rapid environmental changes could pose significant safety risks. This past research has shown that supervised and unsupervised learning models are useful in anticipating emergency scenarios and adapting communication strategies to prioritize safety and reliability.

(d). Multi-Agent Reinforcement Learning for Cooperative Air Transportation

Park et al. (2023) examine multi-agent reinforcement learning (MARL) frameworks for cooperation in autonomous UAM systems. Their research centers on city-wide autonomous UAM, where multiple air taxis must coordinate to reduce traffic congestion and avoid communication interference. MARL enables UAM agents to learn and improve their actions collectively, optimizing flight paths and communication channels in response to each other's behavior. This approach not only minimizes collisions but also improves the efficiency of communication networks by reducing congestion on shared communication channels. The researchers showed that by incorporating MARL, air taxis can dynamically adjust routes and communication protocols based on real-time traffic conditions, enhancing overall network resilience and service reliability in urban areas.

(e). OFDM-Based DSRC Communication Systems for Urban Air Mobility

Huong, Chien, and Ha (2023) focus on orthogonal frequency-division multiplexing (OFDM)- based Dedicated Short-Range Communications (DSRC) systems, supported by ML algorithms to enhance signal processing in urban mobility. Given the potential for high signal interference in urban environments, DSRC provides low-latency communication essential for UAM. Their work integrates ML models to optimize OFDM parameters dynamically, reducing interference and improving spectral efficiency. Such systems are valuable for UAM operations where high-speed, low-latency communication is crucial for real-time control and navigation. Their research shows that ML-assisted OFDM-DSRC systems significantly enhance communication reliability, even in areas with high interference from other wireless devices.

(f). Federated Learning for 6G-Enabled Secure Communication

Sirohi et al. (2023) provide a comprehensive survey on the potential of federated learning in 6G-enabled secure communication systems, which could be instrumental for UAM networks. Federated learning allows UAM systems to train shared models across decentralized nodes (i.e., air taxis) without sharing raw data, which preserves data privacy and improves security. This is particularly relevant for UAM air taxis, where secure communication is critical to prevent cyberattacks and data breaches. The authors propose that federated learning can be utilized to create adaptive communication models that respond to security threats in real time, ensuring that UAM systems maintain secure and stable connections without compromising data integrity.

(g). Summary and Future Directions

The reviewed literature underscores the transformative potential of machine learning in enhancing communication reliability, performance, and security for urban air mobility systems. Predictive handovers, GNSS performance prediction, MARL for cooperative routing, and federated learning for secure communications represent key areas where ML can address unique UAM challenges. Future research could focus on integrating these machine learning models into a unified communication framework that dynamically adjusts based on real-time conditions and scales efficiently across expanding UAM networks.

3. Autonomous navigation in complex urban environments

Urban Air Mobility (UAM) aims to establish efficient, autonomous, and safe air travel solutions within urban areas. A central challenge for UAM air taxis lies in navigating complex urban landscapes with high-precision, real-time decision-making for obstacle avoidance, trajectory optimization, and route planning. Deep learning and reinforcement learning play critical roles in achieving the accuracy and efficiency required for UAM navigation systems. The following review examines recent advances in deep learning and reinforcement learning applications in UAM navigation, with an emphasis on object detection, trajectory planning, and autonomous guidance.

(a). Object Detection for UAVs

Tang et al. (2024) provided a comprehensive survey on deep learning-based object detection for UAVs, exploring various models and techniques that enhance the detection of objects in complex environments. Object detection is vital for UAM air taxis to ensure safe navigation in urban areas where dynamic obstacles like buildings, vehicles, and other aircraft can affect routes. The authors reviewed convolutional neural networks (CNNs) and region-based CNNs (RCNNs), widely used in UAV applications for their accuracy and computational efficiency. They emphasize that real-time object detection models such as YOLO (You Only Look Once) and SSD (Single Shot MultiBox Detector) are especially suited for high-speed UAM systems, enabling accurate object detection while minimizing processing delays. This technology is critical in UAM applications for obstacle avoidance and in-flight hazard detection.

(b). UAV Trajectory Optimization

Rao et al. (2023) discussed real-time trajectory optimization for UAVs, incorporating potential field approaches to dynamically adapt flight paths in response to nearby obstacles. This approach is highly applicable to UAM air taxis, which require responsive and adaptable navigation systems. Trajectory optimization methods based on potential fields work by treating obstacles as "repulsive forces" and destinations as "attractive forces," guiding UAVs toward safe, optimal paths in real time. Rao et al. propose integrating machine learning techniques into this framework to predict potential obstacles and adjust flight paths proactively. The authors showed that for UAMs, this duo of real-time trajectory planning and machine learning provides efficient and adaptable solutions for navigating urban spaces with dense obstacle configurations.

(c). Deep Learning in UAV Remote Sensing

Osco et al. (2023) reviewed deep learning methods applied in UAV remote sensing, focusing on data acquisition, environmental monitoring, and situational awareness. For UAM systems, remote sensing data enhances situational awareness and allows air taxis to recognize environmental features such as terrain, buildings, and weather conditions that could impact navigation. Osco et al. discuss deep learning models like autoencoders and generative adversarial networks (GANs) that can process and analyze large amounts of aerial imagery, enabling UAVs to detect and interpret environmental patterns. According to the authors, this capability is essential in UAM for pre-flight planning and real-time navigation adjustments, particularly when weather or environmental conditions change rapidly in urban areas.

(d). Proximal Policy Optimization for Autonomous Guidance

Hu et al. (2023) explored the use of proximal policy optimization (PPO), a reinforcement learning algorithm, for autonomous guidance of multi-rotor UAVs. PPO is a robust reinforcement learning technique suitable for continuous action spaces, making it well-suited for UAM applications where air taxis require smooth and adaptive control over flight trajectories. Hu et al. demonstrate that PPO can improve the decision-making processes of UAVs in dynamic environments, allowing them to autonomously adjust their paths in real time based on environmental changes and obstacles. In the UAM context, PPO could be used to train air taxis to handle complex, unpredictable urban scenarios with minimal human intervention, enhancing both autonomy and safety.

(e). Deep Reinforcement Learning for Urban Navigation

Tonti et al. (2025) presented a deep reinforcement learning (DRL) algorithm designed for UAV navigation through simplified urban flow environments. Their research highlights the ability of DRL models to train UAVs to navigate complex, obstacle-laden urban areas by optimizing their routes based on reward functions. By training UAVs in simulated urban environments, Tonti et al. show that DRL can equip UAM systems with the ability to "learn" optimal navigation strategies for different urban layouts. This application of DRL is significant for UAM systems, as it allows air taxis to adapt to and navigate within varied and complex urban infrastructures, improving their operational reliability and safety.

(f). Vision-Based Deep Reinforcement Learning for Obstacle Avoidance

Vision-based DRL algorithms offer robust solutions for obstacle avoidance in UAM navigation. One recent approach integrates computer vision with reinforcement learning to enable UAVs to detect and avoid obstacles in real time. Vision-based DRL systems use camera inputs and deep neural networks to interpret visual data, which is then processed to make navigation decisions. This is especially beneficial for UAM, as air taxis operate in dense urban spaces where obstacles may emerge suddenly. By using vision-based DRL, UAM systems can react quickly to unexpected hazards, optimizing their navigation strategies for safer urban flight.

(g). Memory-Based Deep Reinforcement Learning for Obstacle Avoidance

Singla, Padakandla, and Bhatnagar (2021) examine memory-based DRL for UAV obstacle avoidance in scenarios with limited environmental knowledge. This approach incorporates memory mechanisms, allowing UAVs to "remember" past experiences and improve their navigation strategies. Memory-based DRL is particularly advantageous for UAM systems operating in urban environments where GPS signals may be unreliable or where UAVs might encounter recurring obstacles. By utilizing memory, UAM air taxis can navigate complex cityscapes more effectively, leveraging past information to enhance obstacle avoidance capabilities in real time. The authors demonstrate that memory-based DRL significantly improves navigation performance, particularly in environments with frequent obstacles or limited visibility.

(h). Summary and Future Directions

This literature review (summarized in Table 1) highlights the critical role of machine learning, specifically deep learning and reinforcement learning, in advancing UAM navigation. Object detection, trajectory optimization, and obstacle avoidance are crucial capabilities for UAM air taxis operating in urban areas, and ML-based approaches are well-suited to meet these demands. Future research could focus on integrating these ML models into cohesive navigation frameworks capable of adapting to diverse urban landscapes and handling real-world challenges, such as high-frequency communications, dynamic obstacle management, and safe real-time navigation.

Table 1: Summary of Literature Review on Navigation

Paper Title	**Focus Area**	**Methods Used**	**Key Findings**	**Limitations**
Survey of object detection for UAVs using deep learning (Tang et al., 2024)	Object detection in UAVs	Deep learning methods, including RCNN, SSD, YOLO	Summarizes deep-learning-based object detection methods and their applications in UAVs	Focused mainly on image quality and background complexity, but limited to UAV.
An integrated real-time UAV trajectory optimization and potential field approach (Rao, et al., 2023)	Trajectory optimization, collision avoidance	Chebyshev pseudo-spectral method, Artificial Potential Field	Proposes a real-time trajectory optimization approach combining APF for UAV navigation	Limited to UAV trajectory planning in indoor environments, lacks external environmental factors.
A review on deep learning in UAV remote sensing (Osco et al., 2023)	DL in UAV remote sensing	Deep learning, DNNs, image classification, segmentation	Examines DL applications in UAV remote sensing organizes the literature by application area	Limited focus on demand prediction and navigation for urban air mobility applications
Proximal policy optimization for multi-rotor UAV autonomous guidance (Duoxiu et al., 2022)	Autonomous guidance and obstacle avoidance	PPO, LSTM, Markov decision process, real-time position calculation	Demonstrates improved guidance and obstacle avoidance through reinforcement learning for UAVs	Primarily aimed at multi-rotor UAV tracking, not focused on urban air mobility.
Navigation in a simplified urban flow through deep reinforcement learning (Tonti et al., 2025)	UAV navigation in urban environments	DRL, Proximal Policy Optimization (PPO) with LSTM, fluid flow simulations	The high success rate in navigating UAVs through complex urban environments, minimizing crash rates	Assumes a simplified urban environment
A vision-based deep reinforcement learning algorithm for UAV obstacle avoidance (Roghair et al., 2021)	Obstacle avoidance in UAVs	Domain Network and Gaussian Deep QNetwork (DQN), - Dueling Double Deep Q Network (D3QN), - Gaussian Mixture Model (GMM), - Domain Network (CNNbased)	Guidance-based exploration predicts the next state and uses a Bayesian Gaussian Mixture Model to guide exploration more efficiently than traditional DQN	
Memory-based deep reinforcement learning for obstacle avoidance in UAV (Singla et al., 2021)	Obstacle Avoidance in UAVs: Deep Reinforcement Learning	Recurrent Neural Networks with Temporal Attention	Demonstrates improved obstacle avoidance in unstructured indoor environments, using the memory of past observations.	Lacks depth information from more advanced sensors

4. Demand Prediction for Air Taxi Services

The success of Urban Air Mobility (UAM) relies heavily on understanding and predicting demand patterns to ensure operational efficiency and scalability. Accurate demand prediction enables optimal resource allocation, improved passenger experience, and enhanced economic viability. This section examines key advancements in machine learning applications for demand forecasting within the UAM context.

(a). Machine Learning for Demand Forecasting in Urban Aviation

Rajendran, Srinivas, and Grimshaw (2021) investigate the use of ML algorithms such as gradient boosting and neural networks to predict demand for air taxi services. Their study highlights the importance of incorporating diverse datasets, including socioeconomic factors, weather patterns, and historical travel data, to achieve accurate predictions. By leveraging ML models, UAM operators can anticipate peak demand periods and allocate air taxis accordingly, minimizing passenger wait times and maximizing fleet utilization.

(b). Predicting Passenger Preferences

Ramasamy et al. (2023) explore the role of ML in understanding passenger preferences for UAM services. By analyzing demographic data, travel habits, and feedback, their models identify key factors influencing demand, such as pricing, travel time, and accessibility. This insight enables personalized service offerings and strategic pricing models, fostering greater adoption of UAM.

(c). Time-Series Analysis for Demand Prediction

Choudhury et al. (2023) apply time-series analysis techniques, including Long Short-Term Memory (LSTM) networks, to model demand trends for UAM services. Their approach captures temporal dependencies in data, providing accurate short-term and long-term forecasts. For UAM systems, this capability is vital for dynamic scheduling and adapting to changing urban mobility patterns.

(d). Geographic Demand Clustering

Patil and Wang (2023) propose a clustering-based framework for identifying high-demand regions for UAM services. Using spatial data and population density metrics, their ML models group geographic areas with similar demand patterns. This geographic segmentation aids route planning and station placement, enhancing service accessibility and efficiency.

(e). Summary and Future Directions

Demand prediction is a critical component of UAM planning and operations (Ahmed et al., 2020; Eker et al., 2022). Advancements in ML techniques, including gradient boosting, time-series analysis, and clustering, provide robust tools for forecasting demand. Future research should focus on integrating real-time data sources, such as social media trends and urban event schedules, to further enhance predictive accuracy and operational adaptability.

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